

Sagar Bhatiya

B.Tech. Computer Science Engineering

Graphic Era Hill University, Bhimtal

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Professional Summary

Computer Science Engineering student with a CGPA of **8.4/10** and strong foundations in Data Structures and Algorithms (**500+ problems solved**), Artificial Intelligence, and Full-Stack Web Development. Experienced in developing secure transaction workflows, REST APIs, custom Retrieval-Augmented Generation (RAG) pipelines, and database-driven web systems. Seeking a Software Engineer role to apply analytical problem solving, system design, and AI development skills to real-world software engineering challenges.

Education

Graphic Era Hill University, Bhimtal

Bachelor of Technology in Computer Science Engineering

2023 – 2027

CGPA: 8.4/10

Technical Skills

Programming: C++, JavaScript, Python, Java (Basic)

Frontend & Web: React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend & APIs: Node.js, Express.js, REST APIs

Databases & Vector: SQL, MongoDB, ChromaDB, Vector Search, DBMS

AI & LLM Pipelines: Artificial Intelligence, Retrieval-Augmented Generation (RAG), Prompt Engineering, Groq / OpenAI APIs

Security & Systems: JWT Authentication, Hybrid Encryption, Idempotency, Replay Attack Prevention (TTL)

Core CS: Data Structures and Algorithms, OOP, Operating Systems, Computer Networks, Software Engineering

Tools: Git, GitHub, VS Code, Postman, Jupyter Notebooks

Projects

Offline Digital Payment System

Next.js, JavaScript, Bluetooth, Cryptography

- Designed an offline payment architecture enabling secure peer-to-peer transactions without continuous internet connectivity.
- Implemented hybrid encryption using symmetric and asymmetric cryptography to protect transaction data and secure key exchange.
- Applied idempotency mechanisms to prevent duplicate transaction processing and maintain transaction consistency.
- Designed TTL-based transaction validation to mitigate replay attacks by rejecting expired transaction requests.
- Developed an offline-first transaction workflow using device-to-device communication, local transaction handling, and background synchronization.

Personal Finance Tracker

MERN Stack, MongoDB, Express.js, React.js, Node.js, REST APIs

- Developed a full-stack application for recording, categorizing, and analyzing personal income and expense transactions.
- Built RESTful APIs with Node.js and Express.js for transaction CRUD operations and integrated MongoDB for persistent data storage.
- Implemented JWT-based authentication and authorization for secure user access control.
- Developed a responsive React.js user interface with Context API for centralized application state management.

AI Candidate Assistant & RAG System

Python, FastAPI, Groq API, ChromaDB / Qdrant, React

- Built an interactive ChatGPT-style recruiter AI assistant integrated with custom Retrieval-Augmented Generation (RAG) pipelines.
- Developed Server-Sent Events (SSE) streaming APIs in FastAPI for real-time token generation and multi-model switching.
- Integrated vector search databases to query candidate resume records and provide accurate, context-aware responses.

Achievements & Coding Milestones

- 500+ DSA Problems Solved:** Practiced extensively across LeetCode, GeeksforGeeks, and Codeforces using C++, mastering algorithm optimization and complex data structures.
- AI & LLM Pipelines:** Engineered custom RAG architectures, vector database search (ChromaDB/Qdrant), and LLM API integrations.
- Academic Excellence:** Maintained a high academic standing with a **CGPA of 8.4/10** in Computer Science Engineering.
- Competitive Problem Solving:** Regular participant in algorithmic coding contests, focusing on optimal time and space complexity execution.